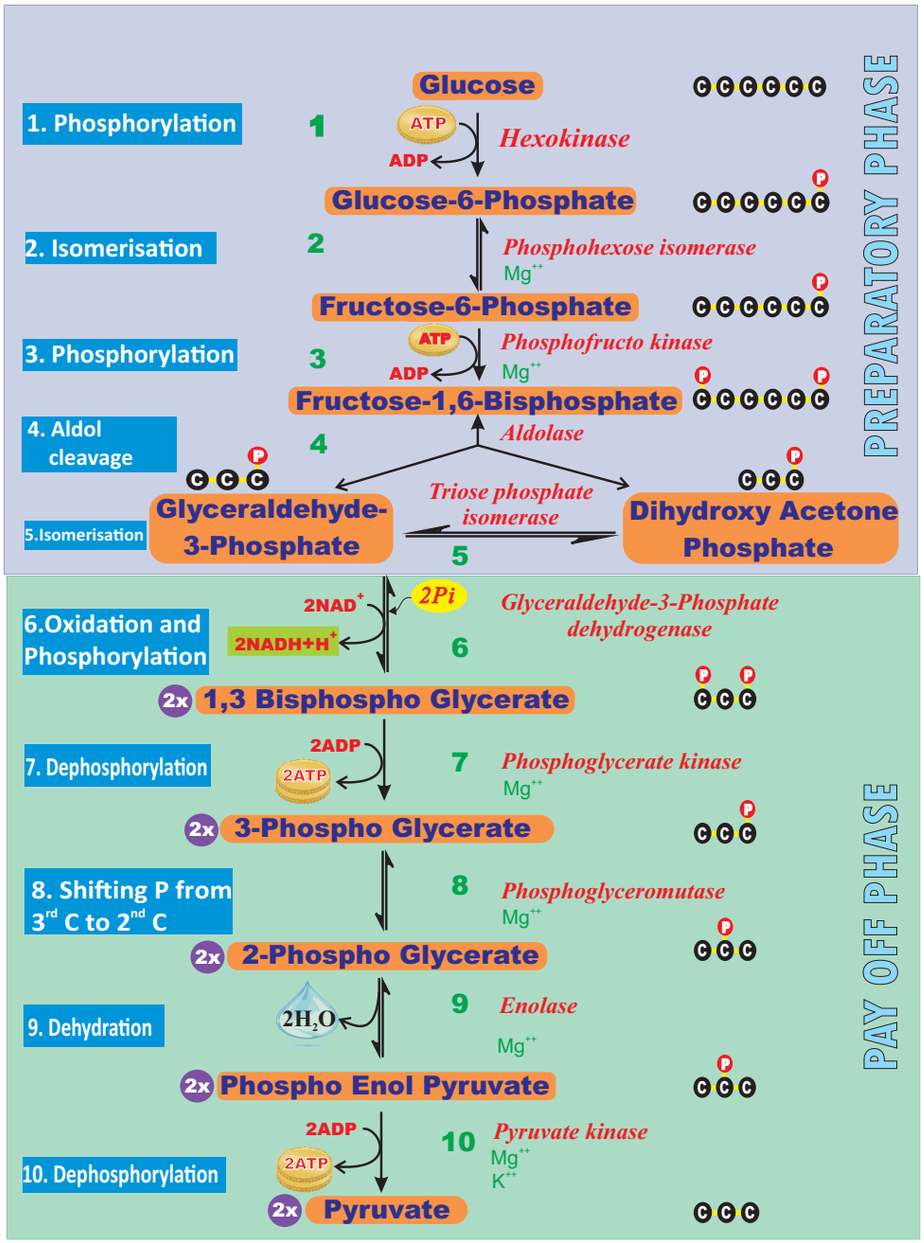


Figure 14.6: Glycolysis or EMP pathway

or **direct phosphorylation** or **trans phosphorylation**. During the reaction at step 9, 2 phospho glycerate dehydrated into Phospho enol pyruvate. A water molecule is removed by the enzyme enolase. As a result, enol group is formed within the molecule. This process is called **Enolation.**

3. Energy Budget

In the pay off phase totally 4ATP and 2NADH + H^+ molecules are produced. Since 2ATP molecules are already consumed in the preparatory phase, the net products in glycolysis are 2ATPs and 2NADH + H^+.

The overall net reaction of glycolysis

$$C_6H_{12}O_6 + 2ADP + 2Pi + 2NAD^+ \rightarrow 2x\ CH_3COCOOH + 2ATP + 2NADH + 2H^+$$

14.5.2 Pyruvate Oxidation (Link reaction)

Two molecules of pyruvate formed by glycolysis in the cytosol enters into the mitochondrial matrix. In aerobic respiration this pyruvate with coenzyme A is oxidatively decarboxylated into acetyl CoA by pyruvate dehydrogenase complex. This reaction is irreversible and produces two molecules of NADH + H^+ and $2CO_2$. It is also called **transition reaction**

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